Geometric Sequences and Series

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Geometric sequences

For the specified parameter configuration, consider the sequence $3, 6, 12, 24, \ldots$

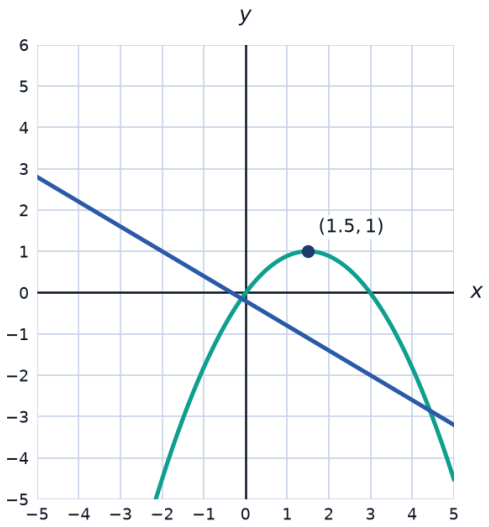
a State the common ratio.

b Write a formula for the nth term.

c Find a_{10}.

2 [Foundational] • Target: Geometric sequences

In a geometric sequence, $a_2 = 6$ and $a_5 = 48$. Find the common ratio and the first term.



3 [Foundational] • Target: Geometric series

Find the sum of the first 8 terms of $5 + 10 + 20 + \cdots$

4 [Foundational] • Target: Geometric series

Evaluate each infinite sum, or state that it diverges:

a $18 + 6 + 2 + \frac{2}{3} + \cdots$

b $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \cdots$

c $2 + 3 + 4.5 + \cdots$

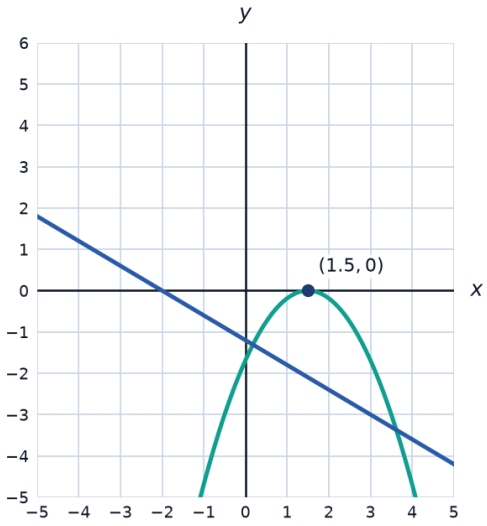
5 [Foundational] • Target: Geometric series

Write the repeating decimal $0.\overline{7}$ as a fraction using an infinite geometric series.

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6 [Foundational] • Target: Geometric series

A ball is dropped from 8 m and rebounds to $\frac{3}{4}$ of its previous height on each bounce. Find the total vertical distance it travels (falling and rising) before coming to rest.



7 [Foundational] • Target: Geometric means and mixed sequences

Insert two positive geometric means between 4 and 108 (i.e., find b_1, b_2 so that $4, b_1, b_2, 108$ is geometric).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Geometric means and mixed sequences

Determine whether each sequence is arithmetic, geometric, or neither:

a $2, 6, 18, 54, \ldots$ **b** $5, 8, 11, 14, \ldots$ **c** $1, 4, 9, 16, \ldots$

9 [Intermediate] • Target: Applications

A car worth 80,000 riyals depreciates by 15% per year.

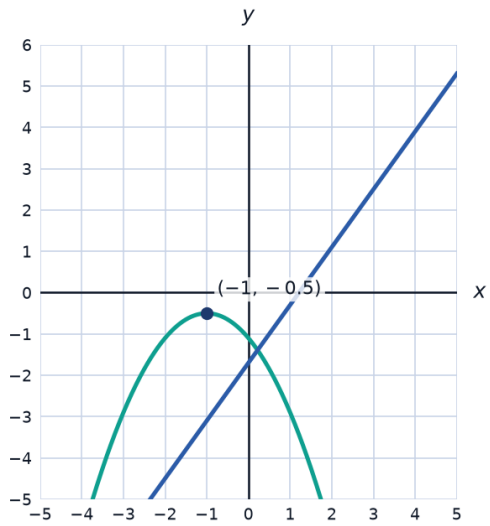
a Write a formula for its value after n years.

b Find its value after 5 years, to the nearest riyal.

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10 [Intermediate] • Target: Applications

A chain letter starts with 1 person sending it to 5 people, each of whom sends it to 5 more, and so on. Find the total number of people who have received the letter after 6 rounds (including the original sender's round).



11 [Intermediate] • Target: Visualizing geometric growth

The first 6 terms of $a_n = 2(1.5)^{n-1}$ are plotted below. Contrast the shape of this graph with the straight-line shape of an arithmetic sequence's graph.

12 [Intermediate] • Target: Determining convergence before summing

For each infinite geometric series, determine whether it converges WITHOUT computing the sum, by checking $|r| < 1$: $\sum_{n=1}^{\infty} 5(0.9)^{n-1}$, $\sum_{n=1}^{\infty} 3(1.2)^{n-1}$, $\sum_{n=1}^{\infty} 7(-0.4)^{n-1}$.

13 [Intermediate] • Target: Finding the number of terms from a partial sum

For the geometric series $2 + 6 + 18 + \cdots$, find the number of terms n needed for the sum to first exceed 2000.

14 [Intermediate] • Target: A mixed application: population with geometric decline

A population of an endangered species declines by 12% each year from an initial 5000. Find the total number of 'animal-years' lived by the population over the first 10 years (i.e. the sum of the population sizes at the start of each of the 10 years), to the nearest whole number.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

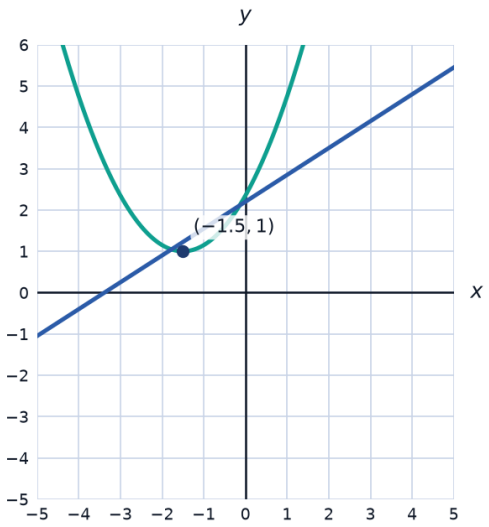
15 [Challenging] • Target: Finding the number of terms from a partial sum • Cross-Lesson: 4.03 Hyperbolas

An infinite geometric series has first term $a_1 = 18$ and sum to infinity $S_\infty = 24$. (a) Find the common ratio r. (b) Calculate the third term a_3.

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16 [Challenging] • Target: A mixed application: population with geometric decline • Cross-Lesson: 5.01 Vectors: Components and Operations

An infinite geometric series has first term $a_1 = 36$ and converges to $S_\infty = 48$. What is the common ratio r, and what is the third term a_3?



17 [Challenging] • Target: Geometric sequences • Cross-Lesson: 5.02 The Dot Product

Evaluate the finite sum: $\sum_{k=1}^{6} 3(2)^{k-1}$

18 [Challenging] • Target: Geometric series • Cross-Lesson: 5.03 Parametric Equations

An infinite geometric series has first term $a_1 = 5$ and common ratio $r = 0.6$. Find its sum S_∞.

19 [Challenging] • Target: Geometric means and mixed sequences • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Find the first three non-zero terms of the Maclaurin series for e^{2x}. State the range of validity.

20 [Challenging] • Target: Applications • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the binomial expansion of $\frac{1}{\sqrt{4-x}}$ in ascending powers of x up to and including the term in x^2. State the set of values of x for which the expansion is valid.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Geometric sequences

How many terms of the geometric series $1 + 3 + 9 + \cdots$ give a sum of 364?

a. 5

b. 7

c. 6

d. 8

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22 [Foundational] • Target: Geometric series

Find the number of terms in an AP with first term 5, last term 41, sum 230.

a. 10

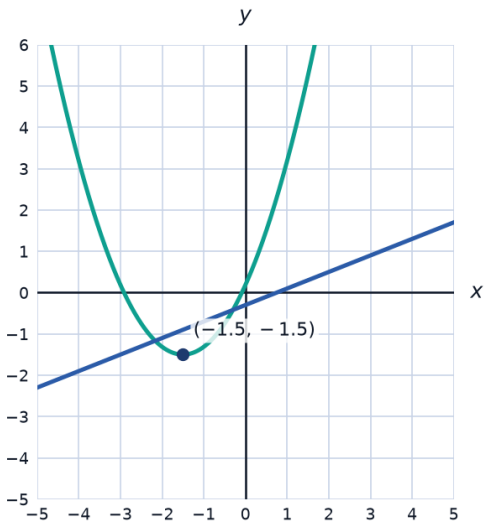
b. 15

c. 12

d. 8

23 [Foundational] • Target: Geometric means and mixed sequences

The sum of the first n terms of a geometric series is $S_n = \frac{a_1(1-r^n)}{1-r}$. Find S_5 for $a_1 = 2, r = 3$.



a. 486

b. 243

c. 242

d. 240

24 [Foundational] • Target: Applications

The first term of a geometric progression is 16 and the fourth term is 2. Find the sum to infinity, S_∞, of the progression.

a. 32

b. $\frac{64}{3}$

c. 24

d. 30

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25 [Foundational] • Target: Visualizing geometric growth

If $|r| < 1$, prove $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ using the limit of partial sums.

a. Use calculus only

b. Cannot be proved

c. $S_N = \frac{1-r^{N+1}}{1-r} \to \frac{1}{1-r}$ as $r^{N+1} \to 0$

d. Only for $0 < r < 1$

26 [Foundational] • Target: Determining convergence before summing

Using partial fractions, find $\sum_{k=1}^{n} \frac{1}{(2k-1)(2k+1)}$.

a. 1

b. $\frac{1}{2}$

c. $\frac{n}{2n-1}$

d. $\frac{n}{2n+1}$

27 [Foundational] • Target: Finding the number of terms from a partial sum

A geometric series with $|r| < 1$ has an infinite sum of:

a. $S = a_1 \cdot r$

b. $S = \frac{a_1}{1-r}$

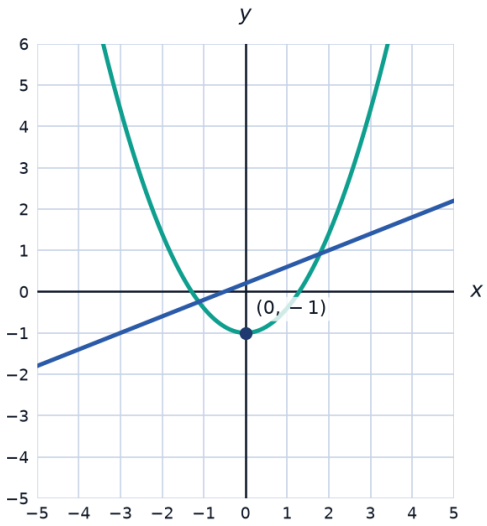
c. Undefined

d. $S = \frac{a_1}{r}$

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28 [Foundational] • Target: A mixed application: population with geometric decline

Use generating functions to find the number of ways to write n as an ordered sum of positive integers.



a. n^2

b. 2^{n-1}

c. $n!$

d. 2^n

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Geometric sequences

Find the sum of an infinite geometric series where $a_1 = 6$ and $r = 2/3$.

a. 6

b. 12

c. 18

d. 9

30 [Intermediate] • Target: Geometric series

A geometric series has $a_1 = 100$ and $r = 0.9$. Find the infinite sum.

a. 900

b. 1000

c. 10

d. 100

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31 [Intermediate] • Target: Geometric means and mixed sequences

What is the sum of the first n terms of $1 + 1 + 1 + \cdots$?

a. n^2

b. n

c. $n/2$

d. ∞

32 [Intermediate] • Target: Applications

For the arithmetic-geometric series $\sum_{k=1}^{\infty} kr^k$ with $|r| < 1$, find the sum.

a. $\frac{r}{(1-r)^2}$

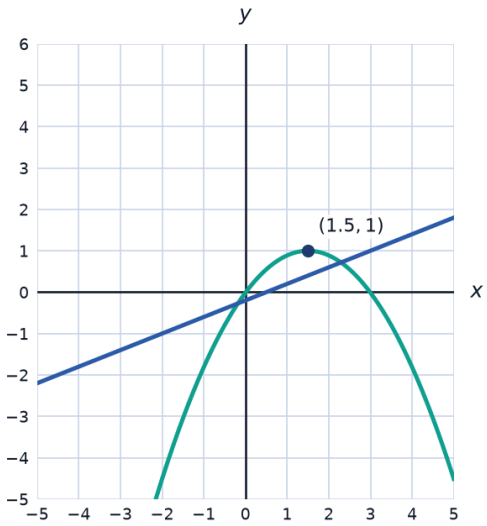
b. $\frac{1}{(1-r)^2}$

c. $\frac{r}{1-r}$

d. $\frac{1}{1-r}$

33 [Intermediate] • Target: Visualizing geometric growth

The geometric series $8 + 4 + 2 + 1 + \cdots$ has infinite sum:



a. 16

b. 32

c. 15

d. 8

34 [Intermediate] • Target: Determining convergence before summing

The nth partial sum of a series is $S_n = n^2 - 3n$. Find the first term a_1.

a. 1

b. 2

c. -2

d. -3

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35 [Intermediate] • Target: Finding the number of terms from a partial sum

Find the sum of the geometric series $1 + \frac{2}{3} + \frac{4}{9} + \cdots$ (infinite).

a. $\frac{3}{2}$

b. 3

c. 2

d. ∞

36 [Intermediate] • Target: A mixed application: population with geometric decline

Find how many terms of the arithmetic series $5 + 8 + 11 + \cdots$ are needed for the sum to exceed 100.

a. $n = 7$

b. $n = 8$

c. $n = 10$

d. $n = 9$

37 [Intermediate] • Target: Geometric sequences

Determine convergence of $\sum_{n=1}^{\infty} \frac{n!}{n^n}$.

a. Converges to e

b. Converges (ratio test: $a_{n+1}/a_n \to 1/e < 1$)

c. Inconclusive

d. Diverges

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Geometric series • Cross-Lesson: 5.03 Parametric Equations

A geometric series has sum $S_n = 5(1 - 0.2^n)$. Identify a_1 and r.

a. $a_1 = 1,\ r = 0.2$

b. $a_1 = 4,\ r = 0.8$

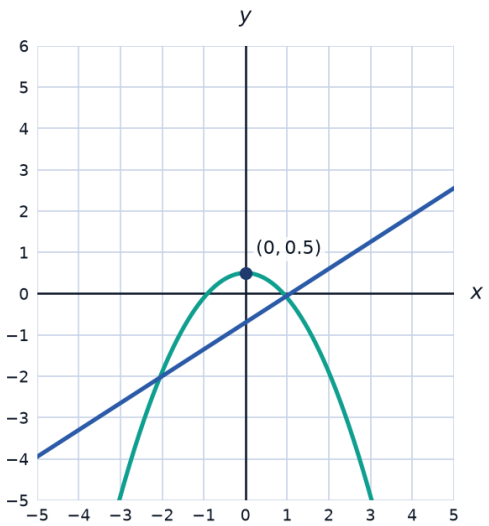
c. $a_1 = 4,\ r = 0.2$

d. $a_1 = 5,\ r = 0.2$

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39 [Challenging] • Target: Geometric means and mixed sequences • Cross-Lesson: 6.01 Polar Coordinates and Conversions

A geometric series has first term $a = 2$ and common ratio $r = -\frac{1}{4}$. Find the sum to infinity.



a. $\frac{8}{5}$

b. $\frac{18}{5}$

c. $\frac{5}{2}$

d. $-\frac{8}{5}$

40 [Challenging] • Target: Applications • Cross-Lesson: 6.02 Polar Equations and Graphs

For the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$, find the partial fraction and sum.

a. ∞

b. $\frac{1}{2}$

c. 1

d. $\ln 2$

41 [Challenging] • Target: Visualizing geometric growth • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

The sum of the first n terms of a series is $S_n = n^2 + 3n$. Is the series arithmetic?

a. Yes, $d = 3$

b. No

c. Yes, $d = 2$

d. Cannot determine

42 [Challenging] • Target: Determining convergence before summing • Cross-Lesson: 7.02 Logarithms and Their Properties

A geometric series has first term $a = 5$ and common ratio $r = 2$. Find S_3, the sum of the first 3 terms.

a. 35

b. 40

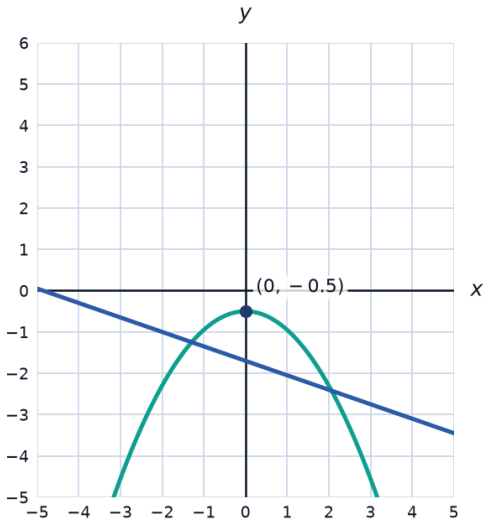
c. 75

d. 15

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43 [Challenging] • Target: Finding the number of terms from a partial sum • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

What is $\sum_{k=0}^{n} r^k$ (geometric sum, $req1$)?



a. $(n+1)r^n$

b. $\frac{r^n-1}{r-1}$

c. r^n

d. $\frac{r^{n+1}-1}{r-1}$

44 [Challenging] • Target: A mixed application: population with geometric decline • Cross-Lesson: 7.04 Applications: Growth and Decay

Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.

a. $R = e$

b. $R = \infty$

c. $R = 0$

d. $R = 1$

45 [Challenging] • Target: Geometric sequences • Cross-Lesson: 8.01 Rational Functions and Asymptotes

A geometric series with common ratio $r = \frac{1}{3}$ has sum to infinity $S_\infty = 12$. Find the first term a.

a. 11

b. 7

c. 8

d. -8

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